DDIS 3D Binning: Purity, Stability, Efficiency

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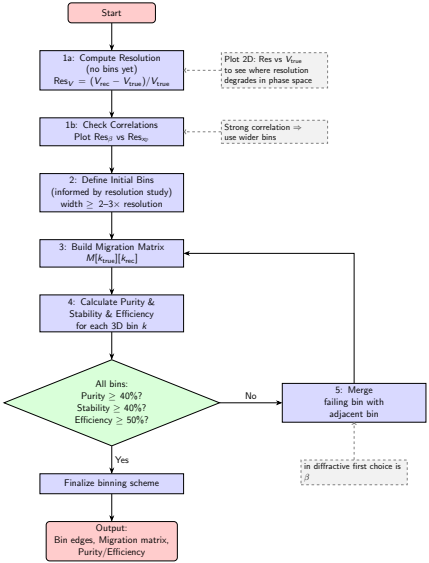
Definitions (Words)

- **Purity**: fraction of events reconstructed in a bin that were generated in the same bin.
- **Stability**: fraction of events generated in a bin that are reconstructed in the same bin.
- **Efficiency**: fraction of events generated in a bin that are reconstructed in any bin (i.e., pass reconstruction).

$$\mathrm{Purity}_k = \frac{N_k^{\mathrm{rec}\wedge\mathrm{true}}}{N_k^{\mathrm{rec}}}, \quad \mathrm{Stability}_k = \frac{N_k^{\mathrm{rec}\wedge\mathrm{true}}}{N_k^{\mathrm{true}}}, \quad \mathrm{Efficiency}_k = \frac{N_k^{\mathrm{rec}}}{N_k^{\mathrm{true}}}$$

3D Binning Flowchart

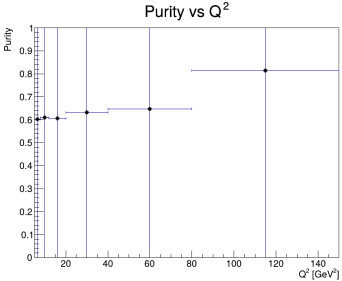


Binning Scheme File (data/binning_scheme.txt)

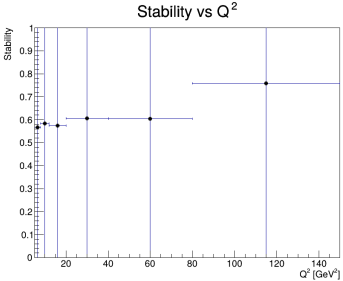
```
# Example binning scheme for DDIS 3D binning
# Hierarchy: Q2 (master), x_pom (secondary), beta (tertiary)

Q2_BINS 6
5.0 8.0 12.0 20.0 40.0 80.0 150.0

BETA_BINS 5
0.0 0.2 0.4 0.6 0.8 1.0

XPOM_BINS 7
1e-4 3e-4 1e-3 3e-3 1e-2 3e-2 1e-1 3e-1
```

The global bin index for a 3D cell (i, j, l) corresponding to $(Q_i^2, x_{\mathbb{P},j}, \beta_l)$ is:
$k = l + j.N_\beta + i.N_\beta.N_{x_\mathbb{P}}$

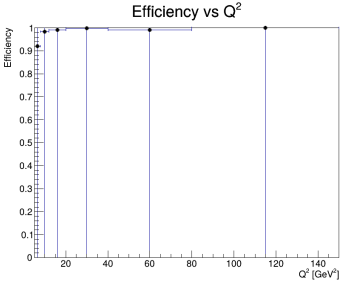
Purity / Stability / Efficiency (Q^2)



Purity vs Q^2



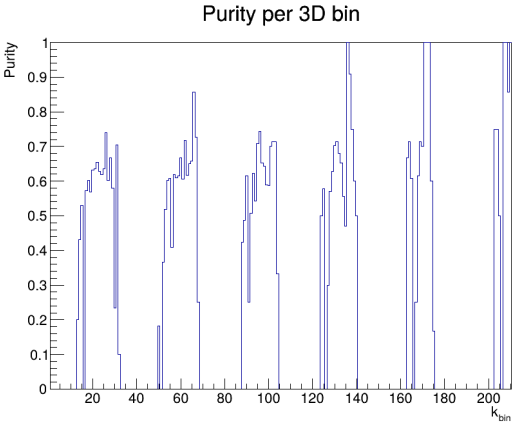
Stability vs Q^2



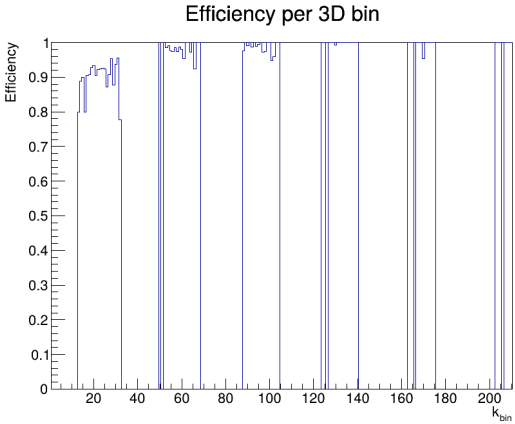
Efficiency vs Q^2

Bins with low efficiency are candidates for merging in the binning workflow.

Purity / Efficiency (All 3D Bins)



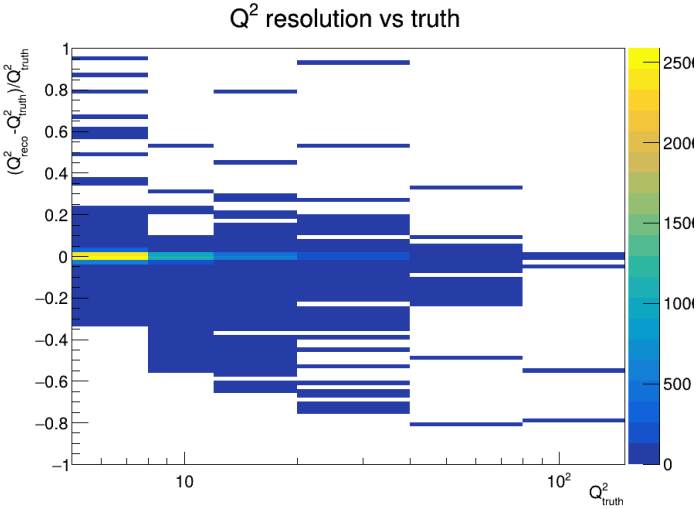
Purity per 3D bin



Efficiency per 3D bin

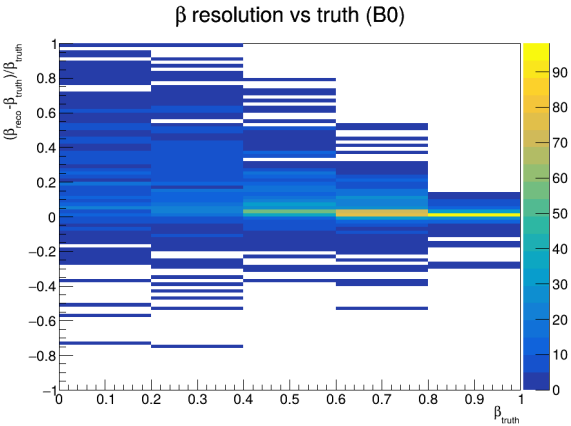
Use these to identify bins failing the target criteria.

Resolution (All Methods)

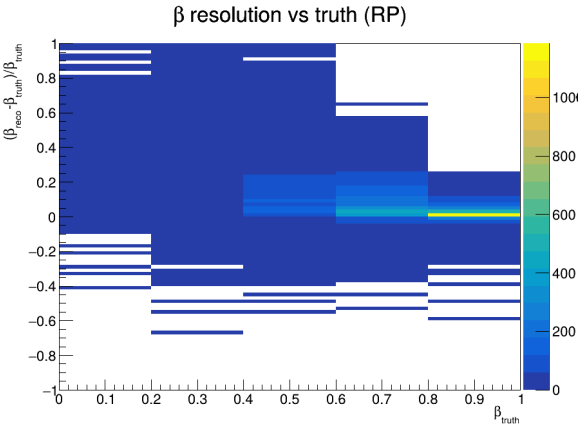


$$\mathrm{Res}_{Q^2} = (Q^2_{\mathrm{rec}} - Q^2_{\mathrm{true}})/Q^2_{\mathrm{true}}$$

Resolution (B0 / RP)

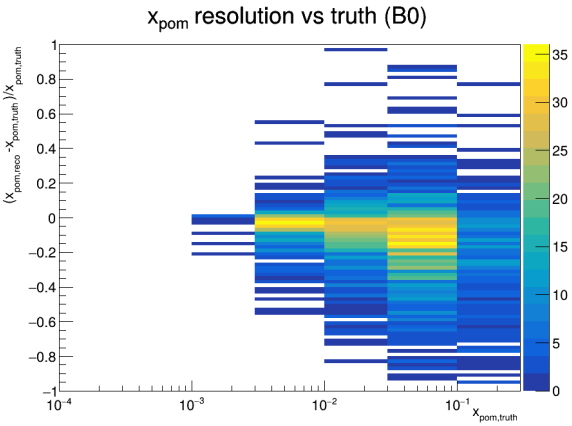


β resolution (B0)

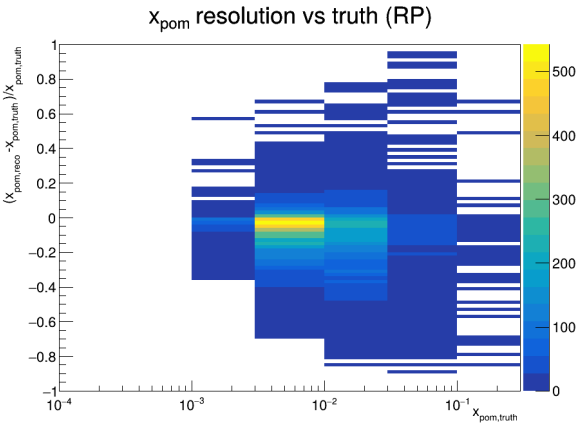


β resolution (RP)

Resolution (B0 / RP)

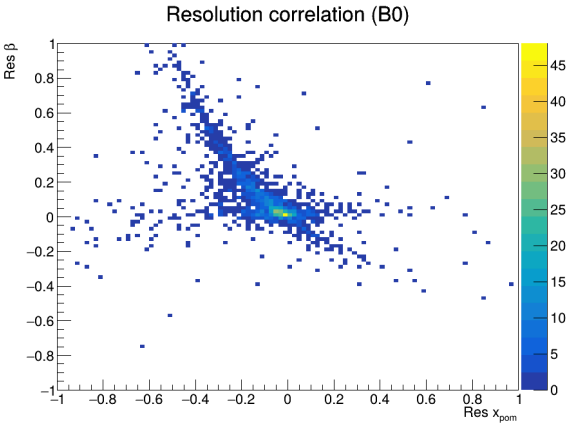


x_{pom} resolution (B0)

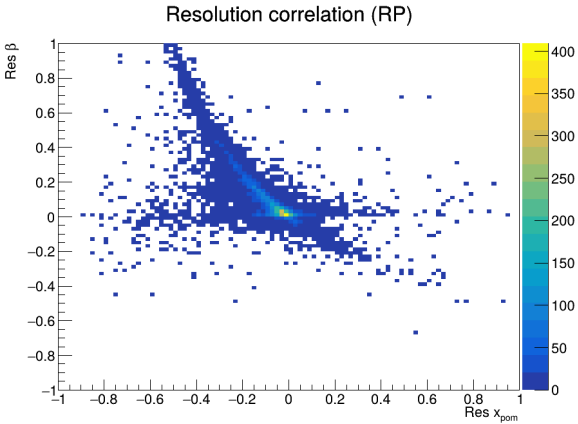


x_{pom} resolution (RP)

Resolution Correlations



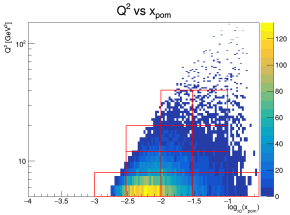
Res(β) vs Res(x_{pom}) (B0)



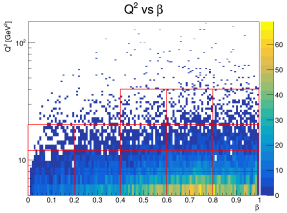
Res(β) vs Res(x_{pom}) (RP)

Correlations guide whether bins should be widened or merged.

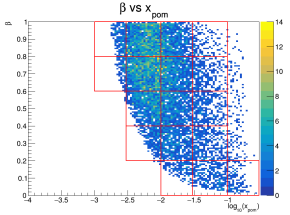
Binning Scheme Overlays



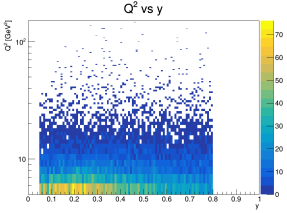
Q^2 vs x_{pom}



Q^2 vs β



β vs x_{pom}



Q^2 vs y